

Open Letter

The general letter - the same form as the individual company letters, without claiming research into any particular company.

To the robotics, automation, AI, and physical-systems teams I meet at RoboBoston: thank you for being here, for sharing what you're building, and for taking the time to consider this.

I am making a deliberate transition into robotics, bringing experience in mechanical troubleshooting, construction and fabrication, routed audio systems, mixed-system maintenance, and electrical study.

In audio work, I traced signal chains through sources, processing, routing and control points, looking for where the observed output differed from what I expected.

Troubleshooting became an exercise in topology. Where did the signal originate? What should happen to it next? What transformed it? What controlled that transformation? Where did the observed behavior first diverge from the expected behavior?

Tracing a fault through connected components is useful in mechanical and venue-system troubleshooting as well as audio.

Mechanical and automotive work taught me to trace physical causality through engines, drivetrains, electrical faults, sensors, and failure modes. Metal fabrication and welding taught me that technical ideas eventually have to survive material reality: fit, access, sequence, tooling, force, heat, repairability, and the difference between what should work and what actually does.

At Level99, my paid Venue Technician work involved troubleshooting and maintaining interactive systems combining RFID and other sensors, PLC-monitored elements, DMX, Arduino hardware, programmable LEDs, audio, mechanical components, and computer control. I used PLC monitoring interfaces to observe states and test sensors; software personnel handled ladder logic. The important part was not any one technology. It was the interaction between them.

At the Starlink facility in Bastrop, Texas, I built clean-room structures for PCB manufacturing. I later completed Martin Electrical School's Journeyman One program, adding formal electrical theory, code, and safety to a technical foundation that had already been developing through mechanical work, fabrication, troubleshooting, and systems thinking.

My current technical project work uses AI assistance. I direct and review work, define acceptance criteria, and use tests and recorded evidence to check results. Signal-path reasoning is useful here for examining inputs, transformations, dependencies, and failures; the project artifacts show what has actually been checked.

I now bring together mechanical troubleshooting, construction and fabrication, routed audio work, Level99 maintenance, and electrical education as I pursue robotics technician work. My subsequent AI-assisted projects add work on requirements, tests, and evidence, with their limits stated alongside the results.

I am not presenting myself as an experienced robotics engineer, controls engineer, embedded-firmware specialist, or someone who has already mastered the entire robotics stack. I have not.

What I can bring immediately is hands-on fabrication and welding experience, mechanical troubleshooting, electrical fundamentals, hands-on troubleshooting of sensor-driven interactive systems, comfort with tools and equipment, signal-path reasoning from audio engineering, software and AI systems thinking, and a strong habit of tracing causes across domain boundaries.

I am interested in roles where that foundation is useful now and where I can continue developing deeper professional competence in robotics hardware, electromechanical integration, test, reliability, controls, prototype build, field support, and systems that have to work outside of a diagram.

If that combination is useful to your team, I would like to talk.

Zachary Neagle